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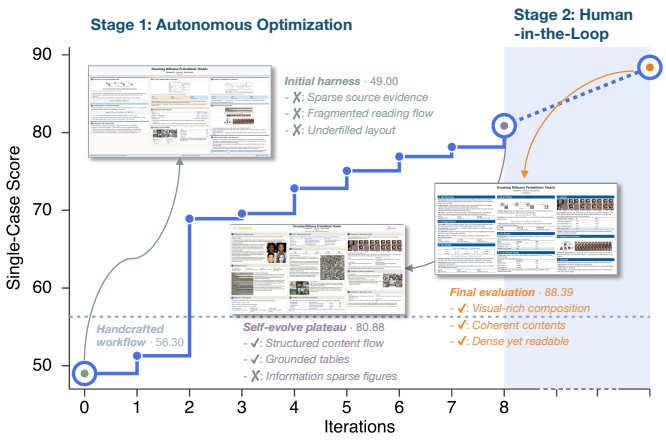
AutoDesign: Meta-Harness Optimization for Long-Horizon Agentic Design

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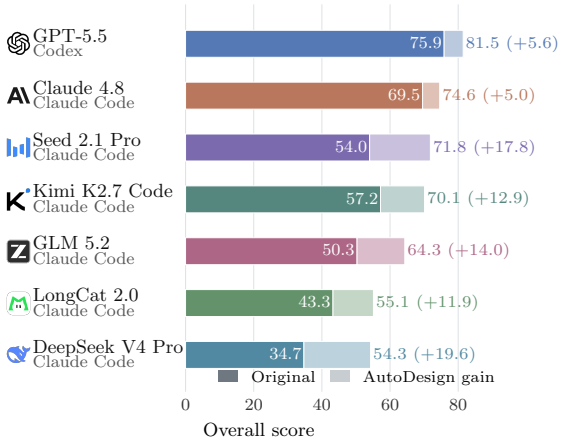
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Transforming multimodal sources into condensed and structured media outputs can be fundamentally conceptualized as a long-horizon agentic process centered on a model-harness system. While an ideal harness system should align with human design priors and accumulate reusable experience through empirical exploration to drive recursive self-improvement, existing paradigms remain static and fall short of this capability. In this paper, we present AutoDesign, a framework that aligns with human design priors, where a meta-harness optimizer guides a code agent to recursively improve harness based on rollout feedback. To instantiate and evaluate this framework, we focus on the academic paper-to-poster generation task and introduce PosterBench, comprising a 100-paper Main Track spanning five disciplines and PosterBench-mini, a shared 10-paper subset for controlled evaluation. On the PosterBench Main Track, AutoDesign achieves the highest score of 78.32, surpassing the closed-source commercial system Claude Design by 7.45 points. Across seven controlled code agent–model configurations, integrating the learned DesignHarness consistently improves performance, increasing the average PosterBench Score from 54.99 to 67.39 (+12.4%). In a fully autonomous long-horizon loop, it executes 253 tool calls and 11 editing turns within 40 minutes for under $3, reaching average conference-poster quality in human evaluation. A system-blind human study further demonstrates that AutoDesign achieves the highest human preference among evaluated systems.

Project Website https://autodesign.designanything.ai/
Code Repository https://github.com/Yaxin9Luo/AutoDesign
Demo Page https://designanything.ai/



(a) Meta-harness optimization trace



(b) Performance gains from DesignHarness

Figure 1 AutoDesign progressively improves the design harness and the quality of the artifacts. (a) Score of the poster generated by the design harness for one representative paper, tracked across meta-harness iterations. Autonomous optimization improves the initial harness before reaching a plateau, after which human guidance redirects the search and yields a further gain. (b) The optimized harness, DesignHarness, improves all Coding Agents on PosterBench by 5.0 to 19.6 points, achieving a best overall score of 81.5.

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Work done during Yaxin's internship at Meituan.

combine multimodal inputs with generation, specialized agents, and visual refinement (Wang et al., 2024; Jaisankar et al., 2025; Pang et al., 2025; Sun et al., 2026; Zhang et al., 2025b; Choi et al., 2026; Vinaykumar et al., 2026). Closely related systems generate slides, webpages, and narrated videos from papers or documents (Fu et al., 2022; Zheng et al., 2025; Ge et al., 2025; Yang et al., 2025; Chen et al., 2025; Zhu et al., 2025). Structured representations such as HTML/CSS and explicit layers further support editability, rendering, and visual inspection (Qu et al., 2025; Liu et al., 2026b; Si et al., 2025; Wu et al., 2024). Together, these works establish the input, output, and representation choices for multimodal output generation, but their run-time feedback generally remains within a fixed production procedure.

Critics, render diagnostics, and regeneration policies can improve the current multimodal output without changing the system that generated it. Self-Refine is the canonical response-level instance: feedback is used to revise the current answer (Madaan et al., 2023). Other agent methods retain experience more persistently: Reflexion stores verbal reflections, Voyager accumulates executable skills, and ExpeL extracts reusable experience from solved tasks (Shinn et al., 2023; Wang et al., 2023; Zhao et al., 2024). These mechanisms preserve useful information beyond a single attempt, but they typically do not update the harness that repeatedly produces outputs. This system-level perspective has early roots in classical accounts of autonomous agents, self-referential learning, and retaining policy changes according to long-term reward effects (Wooldridge and Jennings, 1995; Schmidhuber, 1987; Schmidhuber et al., 1997). These works are conceptual precedents rather than direct algorithms for `AutoDesign`. Recent work separates improvement of model parameters from persistent improvement of the operational scaffold around a fixed model (Ren et al., 2026). TextGrad, DSPy, and GEPA optimize components or declarative pipelines, whereas STOP, GPTSwarm, ADAS, and AFlow search code- or graph-represented workflows (Yuksekgonul et al., 2024; Khattab et al., 2024; Agrawal et al., 2026; Zelikman et al., 2024; Zhuge et al., 2024; Hu et al., 2025; Zhang et al., 2025a). At the full-harness level, A Self-Improving Coding Agent and MOSS update agent source from execution evidence; Meta-Harness, HarnessX, Self-Harness, and Agentic Harness Engineering study searchable harness programs, composable primitives, bounded updates, and outcome attribution (Robeyns et al., 2025; Cai et al., 2026; Lee et al., 2026b; Chen et al., 2026; Zhang et al., 2026a; Lin et al., 2026). HarnessX also uses execution traces as signals for both harness evolution and future model training (Chen et al., 2026). Recursive Harness Self-Improvement specializes a user-constructed, prompt-level multi-agent harness from pairwise revision feedback (Lee et al., 2026a). Gödel Machines provide a proof-based ideal for self-rewriting, while Darwin Gödel Machine and Huxley-Gödel Machine study empirical harness evolution and distinguish immediate performance from future self-improvement potential (Schmidhuber, 2006; Zhang et al., 2026b; Wang et al., 2025). `AutoDesign` instantiates this direction for academic design by evolving a `DesignHarness` for source-grounded, editable artifacts.

Persistent system updates also differ in how they separate learning signals from final evaluation. RHI (Lee et al., 2026a) keeps its evaluation prompt on the evaluator side of the update loop, yet the resulting pairwise history remains a task-local learning signal. Recursive Self-Evolving Agents use an independent development split to gate persistent updates, while Continual Harness, Adaptive Auto-Harness, and Live-SWE-agent study related adaptation settings (Nguyen et al., 2026; Karten et al., 2026; Liu et al., 2026a; Xia et al., 2025). Agent-as-a-Judge further highlights the value of process-level evidence alongside final-outcome assessment (Zhuge et al., 2025). `AutoDesign` uses an independent development acceptance gate for harness updates and evaluates the final `DesignHarness` with `PosterBench` and a system-blind human study.

8 Conclusion

`AutoDesign` turns recurring design failures into improvements to the system that generates future multimodal outputs. Its `MetaHarnessOptimizer` aggregates trajectories, source and rendering diagnostics, evaluator feedback, and reference posters to update one `DesignHarness` component at a time while keeping model weights fixed. This makes paper-to-poster generation a persistent learning process that accumulates design priors and produces editable outputs for direct use or local revision on the Demo Page. We also introduce `PosterBench`, a unified evaluation protocol for academic posters. `DesignHarness` achieves the top score (78.32) and, with Claude Code and Claude 4.8 fixed, surpasses Claude Design by 7.45 points. It also receives the highest Bradley–Terry estimate in a system-blind human study.